

PREDICTING SONG POPULARITY FROM SPOTIFY AUDIO FEATURES: A TEMPORAL ANALYSIS OF CHANGING POPULARITY

Albert Skalinski's Dissertation
DKIT

Research Problem

What makes a Song popular?

Spotify popularity is influenced by:

- Audio characteristics
- Genre
- Release period
- Wider context not present in the dataset

Main issue:

Audio features alone may not fully explain popularity.

Aim and Research Questions

Aim

To investigate whether Spotify audio features can predict song popularity, and whether their importance changes over time.

RQ1

How have the structural determinants of song popularity evolved over time?

RQ2

How accurately can song popularity be predicted using Spotify audio features?

Data and Methods

Dataset

- Kaggle Spotify Music Dataset
- Final cleaned dataset: 4,368 songs
- Target variable: Spotify popularity score
- Main predictors: audio features

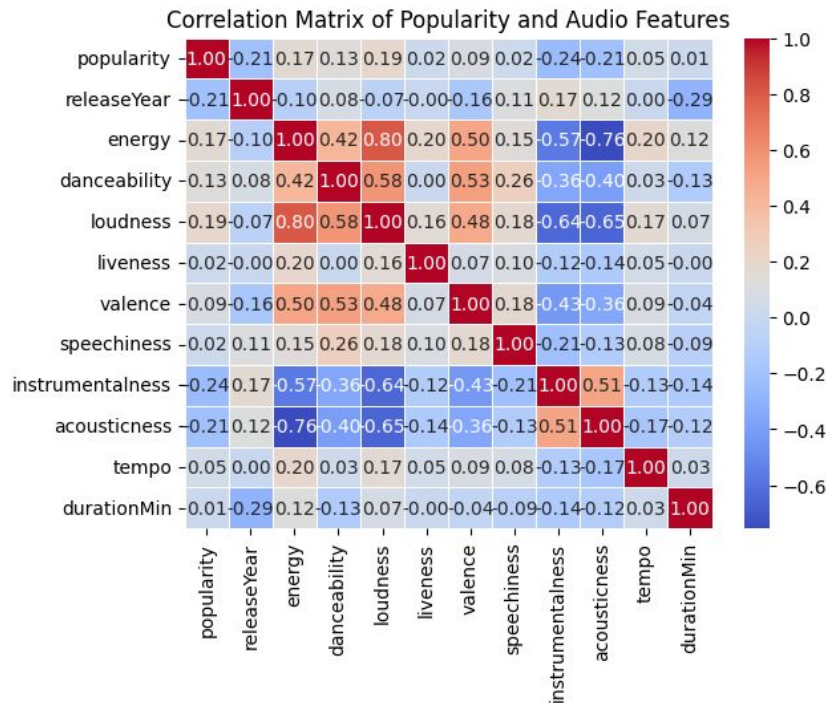
Pipeline

Data cleaning -> EDA -> Cultural/temporal analysis -> Modelling -> Feature importance -> Evaluation

EDA Findings

Key Findings

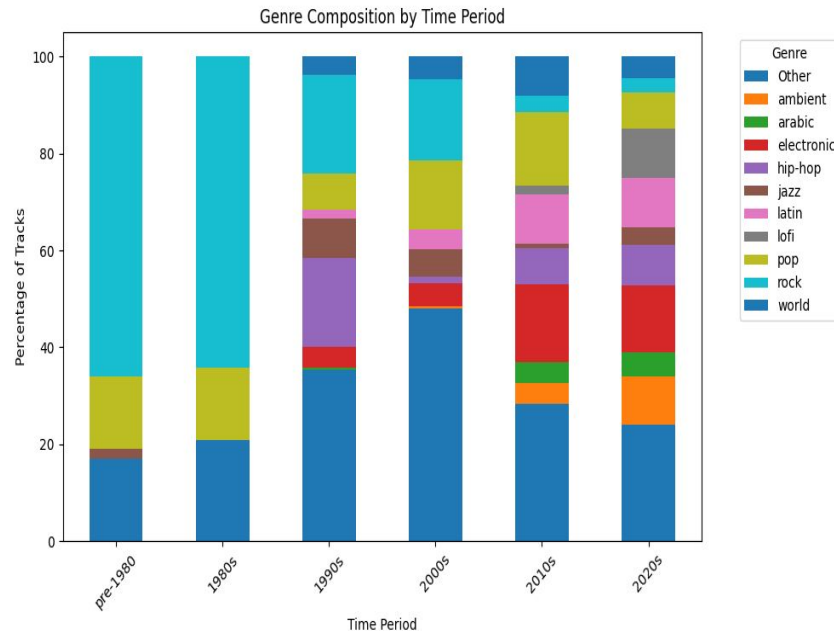
- Popularity distribution was bimodal
- Most songs came from the 2010s and 2020s
- Most audio features had weak linear relationships with popularity
- Acousticness and instrumentality showed clearer negative relationships



Cultural and Temporal Findings

Genre and audio features changed over time

- Older periods were dominated by rock
- Recent periods were more diverse
- Valence and duration decreased over time
- Danceability, speechiness and instrumentality increased



Model Results

Model comparison

Model	R^2
Linear Regression	0.065
XGBoost	0.121
Optimised XGBoost	0.12
Contextual XGBoost	0.342

Main finding

Audio-only models had limited predictive power, but adding genre and release year greatly improved performance.

Conclusion

Final answer

- Spotify audio features can predict popularity, but only to a limited extent
- Genre and release year add important context
- Feature importance changes across time periods
- Older periods: acousticness and valence mattered more
- Recent periods: loudness and danceability mattered more

Limitations

- Imbalanced dataset
- Possible survivorship bias
- No artist popularity, playlist placement, marketing, or stream count data

Thank you for your attention

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